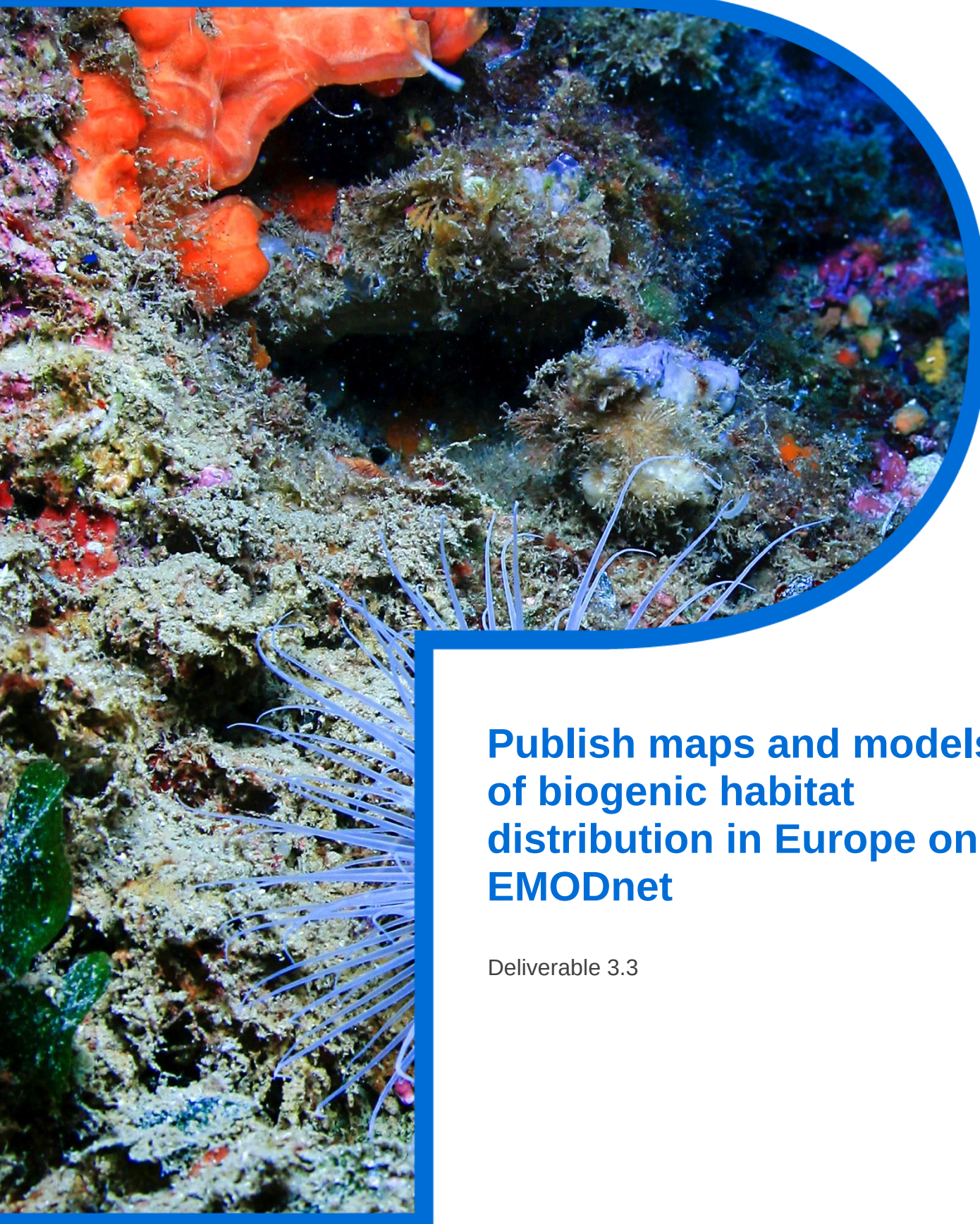




## **Publish maps and models of biogenic habitat distribution in Europe on EMODnet**

Deliverable 3.3



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*The report represents the most up to date status of the information. Any potential upcoming information may be considered for incorporation in future deliverables to improve the scientific knowledge.*



## The MPA Europe project

MPA Europe is a project co-funded by the European Union (EU) that will map a network of locations representative of biodiversity in European seas (Atlantic Exclusive Economic Zones of the EU and its neighbours, and all the Mediterranean, Baltic, and Black seas) (Figure A).



Figure A. MPA Europe study area.

This network will indicate where to prioritise placement of Marine Protected Areas (MPA) for biodiversity protection and how Maritime Spatial Planning (MSP) can maximise blue carbon benefits. By using a holistic set of biodiversity measures, from species to ecosystems (including habitats), and environmental data including carbon storage, water currents, and climate change velocity models, it will be possible to quantify and map both the present and future connectivity within the proposed MPA network. The multiple scenarios provided will support wider and improved MSP by policy makers, industry, and non-governmental organizations (NGOs). The major scientific areas of the project and its outcomes are summarized in the infographic presented in Figure B.

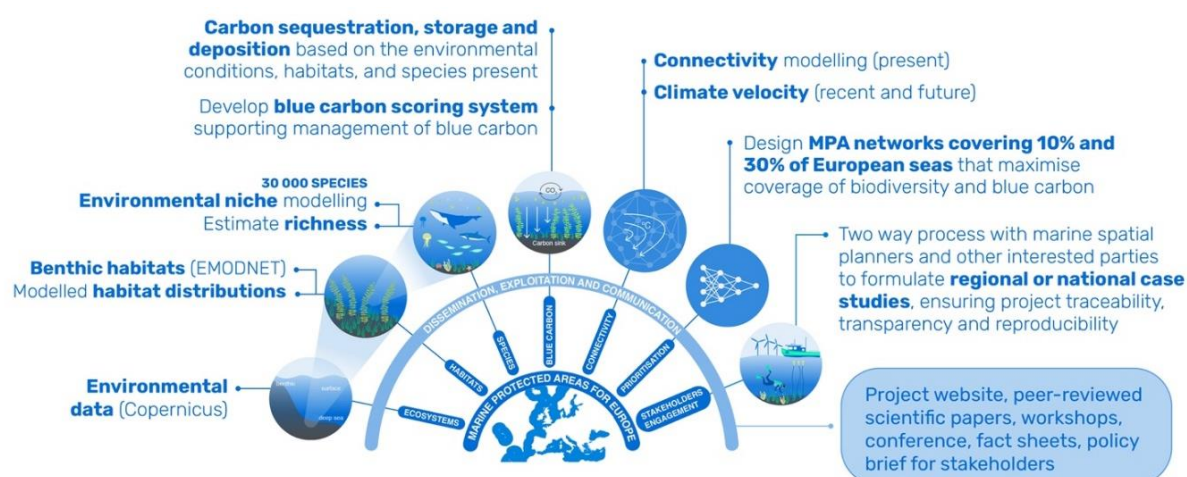


Figure B. MPA Europe project scientific areas

The inter-disciplinary approach of MPA Europe is supported by a diversity of partners. These include Nord University (Norway), the Intergovernmental Oceanographic Commission of UNESCO (Belgium), Aarhus University (Denmark), Climazul (Greece), the Centre of Marine Sciences (Portugal), Science Crunchers (Portugal), the Joint Nature Conservation Committee (England, UK), the Scottish Association for Marine Science (Scotland, UK), the Ocean University of China (China), and the University of the Ryukyus (Japan). The partners include experts in marine biology, taxonomy, ecology, oceanography, biogeochemistry, and biogeography, MPA network design, benthic habitat mapping, carbon dynamics, species distribution, climate modelling, and MSP compose the MPA Europe team.

### **Deliverable 3.3 overview**

This document presents the framework involved in Deliverable 3.3 *Publish maps and models of biogenic habitat distribution in Europe on EMODnet* of MPA Europe. It details the computational procedure, presents maps and the distribution models of biogenic habitats across European seas, considering projected scenarios for both the current and future climate across two time periods (2050 and 2100). Range maps for additional habitat-forming species will become available on the next steps, as data permits. The information provided is crucial for policy-relevant research focusing on MPA designation, in particular, mapping biodiversity patterns, and will feed the main subsequent task of Prioritisation (WP5) of systematic conservation planning to design (prioritise locations for an) MPA networks.

# Predicted changes in the distribution of marine habitat forming species in Europe

## Abstract

Habitat-forming species significantly alter their environment by enhancing its structural complexity, thereby creating resources that support a richer diversity and abundance of species. Consequently, the decline of these biogenic habitats can disrupt cascading effects within ecosystems, leading to impoverished biodiversity. Thus, comprehending the current distribution and potential future changes of these habitats is crucial for conservation efforts. We employed Stacked Species Distribution Models to forecast the distribution of biogenic habitats across European seas, considering nine distinct groups of habitat-forming organisms. Our models projected distributions for both the current and future climate, encompassing five Shared Socio-Economic Pathway scenarios (SSP1, SSP2, SSP3, SSP4, and SSP5) across two time periods (2050 and 2100). Overall, our findings show significant alterations in the composition and extent of biogenic habitats, underscoring the potential impact of climate change on marine ecosystems in Europe. This emphasizes the necessity of implementing Marine Protected Areas that account for the multifaceted aspects of biodiversity.

## Introduction

Multiple groups of species form significant three-dimensional habitat for other species, such as corals (Graham & Nash, 2013; Roberts et al., 2006), macroalgae (Teagle et al., 2017), sponges (Maldonado et al., 2015) and polychaetes (Bonifazi et al., 2019). Due to their structural complexity, these biogenic habitats support a higher diversity and abundance of species (e.g. Graham & Nash, 2013) and the loss of habitat forming species can have cascading effects in the community, including changes in species composition (Alvarez-Filip et al., 2009; Teagle & Smale, 2018). Nevertheless, the identity of habitat forming species is also important, as similar species may have distinct local effects on the structural complexity of the ecosystem. For example, the climate-driven substitution of *Laminaria hyperborea* by its congener *Laminaria ochroleuca* caused a reduction on biodiversity within a temperate community in the south-west United Kingdom (Teagle & Smale, 2018). Thus, understanding changes in the range and composition of biogenic habitats is critical for supporting conservation action.

Worldwide, habitat range maps were produced for seagrass, kelp, and fucoid algae biomes (Assis et al., 2017, 2018; Jayatilake & Costello, 2018, 2020). For Europe, EMODnet (European Marine Observation and Data Network) produces a composite map of seabed habitats (EUSeaMap) which includes biogenic substrata. However, a consistent high-resolution map of biogenic habitat for Europe spanning several groups was still lacking. This is critical, as climate change is already causing negative impacts on habitat forming species along Europe (e.g. Gómez-Gras et al., 2021). Here we apply Stacked Species Distribution Models (SSDM) to derive distributional maps of biogenic habitats for the European seas. To explore the proposed framework, maps were obtained for a selected number of species in nine groups of habitat-forming organisms. Models predicted both the current and future distribution, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 and 2100) according to the most recent models of the CMIP6 (Calvin et al., 2023). The final resulting habitat maps will enable us not only to understand the distribution of biogenic habitats, but also how climate change may impact the occurrence of habitat forming species.

## Methods

### Study area

Habitat distribution maps were produced for all areas within the Exclusive Economic Zone (EEZ) of European Union member states, associated countries, and several Atlantic oceanic islands. For producing the individual species distribution models underlying the habitat maps, we considered an expanded study area ranging from 41°W to 47°E and from 20°N to 89°N to provide enough environmental variability. After model fitting and prediction, all maps were clipped and masked according to the study area of MPA Europe.

### Species distribution models

We employed SSDM to generate habitat distribution maps. This methodology operates on the principle of “predict first, assemble later”, wherein individual SDM are initially fitted and subsequently grouped and aggregated to create richness maps or identify regions where specific groups of species are present (D’Amen et al., 2017; Ferrier & Guisan, 2006). Although alternative methods such as Joint Species Distribution Models (JSDM) exist (Ovaskainen & Abrego, 2020), SSDM are simpler and can even outperform JSDM (Zurell et al., 2020). Furthermore, because our project involves producing individual SDM for at least half of all marine species in European seas (~15,000 species), it is straightforward to produce habitat distribution maps from stacked individual maps.

Our previous report on Deliverable 3.2 (*Publish maps and models of species environmental niches and geographic distributions in Europe on OBIS*) (Principe et al., 2023) extensively covered the framework for our Species Distribution Models (SDM). However, we provide a concise overview of the methodology here. Species occurrence data were sourced from the Ocean Biodiversity Information System (OBIS, 2023) and the Global Biodiversity Information Facility (GBIF, 2023). Automated data cleaning pipelines were deployed to ensure data quality, a step that can significantly impact the outcomes of distribution modelling (Elith & Leathwick, 2009). For each species we further kept only one point per cell (considering the resolution of the environmental variables, ~5 km<sup>2</sup>) and reduced the spatial autocorrelation through an autocorrelation test (following the approach of Boavida et al., 2016).

Environmental datasets were acquired from Bio-ORACLE v3 (Assis et al. 2023), encompassing the current period and five future scenarios based on the Coupled Model Intercomparison Project Phase 6 (CMIP6, <https://wcrp-cmip.org/cmip-phase-6-cmip6/>): SSP1, SSP2, SSP3, SSP4, and SSP5. These datasets included information for the surface (utilized for pelagic species), mean depth (applied to mesopelagic species), and maximum depth (applied to benthic species). Information relative to the mode of life of species, which determined the type of environmental layer being used, were obtained from the databases SeaLifeBase (Palomares & Pauly (Ed.), 2023; <https://www.sealifebase.ca/>), FishBase (Froese & Pauly (Ed.), 2023); <https://www.fishbase.se/>) and WoRMS (Ahyong et al., 2023; <https://www.marinespecies.org/>).

Species distribution models were fitted under a point process approach (Renner et al., 2015) using the methods LASSO (Least Absolute Shrinkage and Selection Operator, a regularized regression; Friedman et al., 2010) and Maxent (Maximum Entropy) which was not used in the previous report on Deliverable 3.2 (*Publish maps and models of species environmental niches and geographic distributions in Europe on OBIS*) (Principe et al. 2023) but was since then incorporated to the framework after adjustments to improve the model fitting time.

Maxent is one of the most used SDM methods, due to its general good performance even in conditions with a low number of occurrence records (Elith et al., 2011; Merow et al., 2013; Phillips & Dudík, 2008). We used the R implementation of Maxent, through the “maxnet” package (Phillips et al., 2017). Because this package uses LASSO under the hood, results were similar between both methods and here we show only the distribution maps obtained with Maxent. Models were evaluated through spatial block-cross validation using the metrics AUC (Area Under the Curve) and CBI (Continuous Boyce Index).

### Habitat forming species

For the purpose of developing the methodology, we selected several species that are known to enhance the structural complexity of the environment and create new resources. These species were categorized into nine distinct groups: habitat forming algae; bryozoan reefs; cold-water coral reefs; coralligenous platforms; deep-sea sponge grounds; mollusc reefs (composed of Gastropoda and Bivalvia species); polychaete reefs (mainly Sabellaridae); seagrass meadows; and shallow-water sponge reefs. Our first list drew upon the literature (e.g. Assis et al., 2020; Balogh et al., 2023; Maldonado et al., 2015).

From the initial list of candidate habitat-forming species, we selected only those species for which we had already produced species distribution maps of sufficient quality (measured by an AUC higher than 0.7) (see Deliverable 3.2, Principe et al., 2023) to use in generating the habitat distribution maps in our study area. It is important to note that several significant habitat-forming species were not incorporated into the habitat maps presented in this report, because they are yet to be modeled and the full modeling of species distribution for this project is underway (Principe et al. 2023). This selection resulted in a list of 61 habitat forming species including algae, bryozoans, cold-water corals, sponges, molluscs, polychaetes, and seagrass species (Table 1).

### Data availability

The codes relative to the modelling of the European species are available at [https://github.com/iobis/mpaeu\\_sdm](https://github.com/iobis/mpaeu_sdm), with an archived version on Zenodo (to be attributed).

*Table 1. List of modeled biogenic habitat distributions. Habitat distribution maps were produced by aggregating individual species distribution models of habitat-forming species obtained through the Maxent algorithm. In total, 61 models included 7 habitat-forming algae, 2 bryozoans, 31 cold-water corals, 2 coralline algae, 3 shallow-water sponges, 6 deep-sea sponges, 2 molluscs, 6 polychaetes, and 2 seagrass species.*

| Biogenic habitat      | Species                        | Phylum              | Class              | Order           | Family           |
|-----------------------|--------------------------------|---------------------|--------------------|-----------------|------------------|
| Habitat-forming algae | <i>Flabellia petiolata</i>     | Chlorophyta         | Ulvophyceae        | Bryopsidales    | Udoteaceae       |
|                       | <i>Halimeda tuna</i>           | Chlorophyta         | Ulvophyceae        | Bryopsidales    | Halimedaceae     |
|                       | <i>Palmophyllum crassum</i>    | Prasinodermatophyta | Palmophyllophyceae | Palmophyllales  | Palmophyllaceae  |
|                       | <i>Peyssonnelia rosamarina</i> | Rhodophyta          | Florideophyceae    | Peyssonneliales | Peyssonneliaceae |
|                       | <i>Peyssonnelia rubra</i>      | Rhodophyta          | Florideophyceae    | Peyssonneliales | Peyssonneliaceae |
|                       | <i>Peyssonnelia squamaria</i>  | Rhodophyta          | Florideophyceae    | Peyssonneliales | Peyssonneliaceae |
|                       | <i>Zanardinia typus</i>        | Ochrophyta          | Phaeophyceae       | Tilopteridales  | Cutleriaceae     |
|                       | <i>Pentapora fascialis</i>     | Bryozoa             | Gymnolaemata       | Cheilostomatida | Bitectiporidae   |

| Biogenic habitat       | Species                           | Phylum   | Class        | Order           | Family           |
|------------------------|-----------------------------------|----------|--------------|-----------------|------------------|
| Bryozoan reefs         | <i>Reteporella grimaldii</i>      | Bryozoa  | Gymnolaemata | Cheilostomatida | Phidoloporidae   |
|                        | <i>Acanella arbuscula</i>         | Cnidaria | Anthozoa     | Scleractyonacea | Keratoisididae   |
|                        | <i>Acanthogorgia hirsuta</i>      | Cnidaria | Anthozoa     | Malacalcyonacea | Paramuriceidae   |
|                        | <i>Alcyonium palmatum</i>         | Cnidaria | Anthozoa     | Malacalcyonacea | Alcyoniidae      |
|                        | <i>Astroides calycularis</i>      | Cnidaria | Anthozoa     | Scleractinia    | Dendrophylliidae |
|                        | <i>Bebryce mollis</i>             | Cnidaria | Anthozoa     | Malacalcyonacea | Paramuriceidae   |
|                        | <i>Callistephanus pallida</i>     | Cnidaria | Anthozoa     | Malacalcyonacea | Gorgoniidae      |
|                        | <i>Callogorgia verticillata</i>   | Cnidaria | Anthozoa     | Scleractyonacea | Primnoidae       |
|                        | <i>Cladocora caespitosa</i>       | Cnidaria | Anthozoa     | Scleractinia    | Cladocoridae     |
|                        | <i>Corallium rubrum</i>           | Cnidaria | Anthozoa     | Scleractyonacea | Coralliidae      |
| Cold-water coral reefs | <i>Dendrophyllia cornigera</i>    | Cnidaria | Anthozoa     | Scleractinia    | Dendrophylliidae |
|                        | <i>Desmophyllum pertusum</i>      | Cnidaria | Anthozoa     | Scleractinia    | Caryophylliidae  |
|                        | <i>Eunicella singularis</i>       | Cnidaria | Anthozoa     | Malacalcyonacea | Eunicellidae     |
|                        | <i>Eunicella verrucosa</i>        | Cnidaria | Anthozoa     | Malacalcyonacea | Eunicellidae     |
|                        | <i>Funiculina quadrangularis</i>  | Cnidaria | Anthozoa     | Scleractyonacea | Funiculinidae    |
|                        | <i>Kophobelemnion stelliferum</i> | Cnidaria | Anthozoa     | Scleractyonacea | Kophobelemnidae  |
|                        | <i>Madracis pharensis</i>         | Cnidaria | Anthozoa     | Scleractinia    | Pocilloporidae   |
|                        | <i>Madrepora oculata</i>          | Cnidaria | Anthozoa     | Scleractinia    | Oculinidae       |
|                        | <i>Oculina patagonica</i>         | Cnidaria | Anthozoa     | Scleractinia    | Oculinidae       |
|                        | <i>Paramuricea clavata</i>        | Cnidaria | Anthozoa     | Malacalcyonacea | Paramuriceidae   |
|                        | <i>Paramuricea macrospina</i>     | Cnidaria | Anthozoa     | Malacalcyonacea | Paramuriceidae   |
|                        | <i>Pennatula aculeata</i>         | Cnidaria | Anthozoa     | Scleractyonacea | Pennatulidae     |
|                        | <i>Pennatula phosphorea</i>       | Cnidaria | Anthozoa     | Scleractyonacea | Pennatulidae     |
|                        | <i>Pennatula rubra</i>            | Cnidaria | Anthozoa     | Scleractyonacea | Pennatulidae     |
|                        | <i>Pteroeides griseum</i>         | Cnidaria | Anthozoa     | Scleractyonacea | Pennatulidae     |
|                        | <i>Savalia savaglia</i>           | Cnidaria | Anthozoa     | Zoantharia      | Parazoanthidae   |
|                        | <i>Solenosmilia variabilis</i>    | Cnidaria | Anthozoa     | Scleractinia    | Caryophylliidae  |
|                        | <i>Stylatula elegans</i>          | Cnidaria | Anthozoa     | Scleractyonacea | Virgulariidae    |
|                        | <i>Umbellula huxleyi</i>          | Cnidaria | Anthozoa     | Scleractyonacea | Umbellulidae     |



| Biogenic habitat           | Species                             | Phylum       | Class           | Order           | Family           |
|----------------------------|-------------------------------------|--------------|-----------------|-----------------|------------------|
| Coralligenous platforms    | <i>Veretillum cynomorium</i>        | Cnidaria     | Anthozoa        | Scleralcyonacea | Veretillidae     |
|                            | <i>Viminella flagellum</i>          | Cnidaria     | Anthozoa        | Scleralcyonacea | Ellisellidae     |
|                            | <i>Virgularia mirabilis</i>         | Cnidaria     | Anthozoa        | Scleralcyonacea | Virgulariidae    |
|                            | <i>Lithophyllum byssoides</i>       | Rhodophyta   | Florideophyceae | Corallinales    | Lithophyllaceae  |
|                            | <i>Mesophyllum expansum</i>         | Rhodophyta   | Florideophyceae | Corallinales    | Mesophyllumaceae |
| Deep-sea sponge grounds    | <i>Geodia barretti</i>              | Porifera     | Demospongiae    | Tetractinellida | Geodiidae        |
|                            | <i>Geodia macandrewii</i>           | Porifera     | Demospongiae    | Tetractinellida | Geodiidae        |
|                            | <i>Phakellia robusta</i>            | Porifera     | Demospongiae    | Bubarida        | Bubaridae        |
|                            | <i>Phakellia ventilabrum</i>        | Porifera     | Demospongiae    | Bubarida        | Bubaridae        |
|                            | <i>Tentorium semisuberites</i>      | Porifera     | Demospongiae    | Polymastiida    | Polymastiidae    |
|                            | <i>Thenia muricata</i>              | Porifera     | Demospongiae    | Tetractinellida | Theneidae        |
| Shallow-water sponge reefs | <i>Chondrilla nucula</i>            | Porifera     | Demospongiae    | Chondrillida    | Chondrillidae    |
|                            | <i>Phorbas tenacior</i>             | Porifera     | Demospongiae    | Poecilosclerida | Hymedesmiidae    |
|                            | <i>Spirastrella cunctatrix</i>      | Porifera     | Demospongiae    | Clionaida       | Spirastrellidae  |
| Mollusc reefs              | <i>Hiatella arctica</i>             | Mollusca     | Bivalvia        | Adapedonta      | Hiatellidae      |
|                            | <i>Vermetus triquetrus</i>          | Mollusca     | Gastropoda      | Littorinimorpha | Vermetidae       |
| Polychaete reefs           | <i>Ficopomatus enigmaticus</i>      | Annelida     | Polychaeta      | Sabellida       | Serpulidae       |
|                            | <i>Protula intestinum</i>           | Annelida     | Polychaeta      | Sabellida       | Serpulidae       |
|                            | <i>Sabellaria alveolata</i>         | Annelida     | Polychaeta      |                 | Sabellariidae    |
|                            | <i>Sabellaria spinulosa</i>         | Annelida     | Polychaeta      |                 | Sabellariidae    |
|                            | <i>Serpula vermicularis</i>         | Annelida     | Polychaeta      | Sabellida       | Serpulidae       |
|                            | <i>Vermiliopsis infundibulum</i>    | Annelida     | Polychaeta      | Sabellida       | Serpulidae       |
| Seagrass meadows           | <i>Posidonia oceanica</i>           | Tracheophyta | Magnoliopsida   | Alismatales     | Posidoniaceae    |
|                            | <i>Zostera subg. Zostera marina</i> | Tracheophyta | Magnoliopsida   | Alismatales     | Zosteraceae      |

### Biogenic habitat distribution maps

The distribution maps for individual species were converted into presence-absence (binary) maps using the 10<sup>th</sup> percentile training presence threshold, referred to as P10. This threshold defines the potential occupancy area of a species as regions where the suitability values exceed those found in the 10% occurrence records with the least suitability values. We chose a metric-independent threshold to convert continuous SDMs to binary maps because our models are based on presence-only information, rather than presence-absence data. Under these circumstances metrics like AUC

and TSS (True Skill Statistic) can be sensitive to species prevalence (Grimmett et al., 2020), consequently influencing the thresholding process.

The binary maps were aggregated based on groups and then summed to generate the habitat-forming composite map for each group. These maps illustrate (1) the geographical distribution of habitat-forming groups, considering the areas with at least one species, and (2) the number of species belonging to each group in different regions (richness). It is important to note that richness maps derived from SDMs may sometimes overpredict or underpredict the number of species in the most extreme areas (Calabrese et al., 2014).

All models and analyses were run in R version 4.3.1 (R Core Team). Raster layers of the habitat distribution were created using the packages “terra” and “sf”, and final maps were produced using “ggplot2”. Habitat distribution maps are in process of publishing in the EMODnet platform (<https://www.emodnet-ingestion.eu/>). The code relative to the modelling of the European species, including the code used to produce the habitat distribution maps, is available at [https://github.com/iobis/mpaeu\\_sdm](https://github.com/iobis/mpaeu_sdm), with an archived version on Zenodo (to be attributed). Methods applied here are also being documented on the MPA Europe documentation, available at [https://iobis.github.io/mpaeu\\_docs/](https://iobis.github.io/mpaeu_docs/).

## Results and Discussion

The outcomes from the applied models revealed that climate change may lead to changes in the composition of habitat-forming species within Europe. This is evident for cold-water coral reefs (Figure 15 and Figure 16), coralligenous platforms (Figure 11 and Figure 12), habitat-forming algae (Figure 1 and Figure 2) as well as for deep-sea sponge grounds (Figure 7 and Figure 8) and polychaete reefs (Figure 9 and Figure 10). While the latter two groups may also experience a reduction of habitat area these are preliminary results on a sample of species (Table 2).

For other groups it is hard to trace this trend as we could only include a few habitat-forming species. For example, molluscs comprise only two species, and although there is a predicted expansion (Table 2), it is also evident that the space where both species co-occur will be reduced (Figure 3 and Figure 4). Seagrass meadows (Figure 17 and Figure 18), and shallow-water sponge reefs (Figure 5 and Figure 6) all show a reduction in area. Bryozoan reefs, on the other hand, may be the habitat with largest expansion, especially on the worst-case scenarios (Table 2, Figure 13 and Figure 14). Within the macroalgae group, forest-forming species were not included in this first map, and its further consideration will provide a better overview of this habitat-forming group.

Although the results obtained in this study indicate certain general trends, it is important to note that several significant species were not incorporated into the habitat maps presented in this report. For instance, range models of key macroalgae species such as *Macrocystis pirifera* and *Laminaria* species, which are considered major habitat formers (Miller et al., 2018; Teagle et al., 2017; Teagle & Smale, 2018), are in preparation (Principe et al. 2023). Similarly, molluscs like *Mytilus edulis* and *Ostrea edulis* were absent from this first modeling realization. Models for these species are being developed as part of the next phase of the project, utilizing updated methodologies.

### Next steps

On closer examination of the species reported as habitat-forming in the literature, we notice that many authors include any sessile and sedentary epifaunal species as habitat-forming, or use higher taxonomic levels to include species when only a subset of the species form significant habitat. Based on this approach one could argue many infaunal species also form biogenic habitat through their

burrows and tubes, and any sedentary species forms habitat for some species. Another issue is that EUNIS also has its own typology for what it classifies as “biogenic habitats” that differs from some use in the literature and this needs to be considered. One of the species here, *Undaria pinnatifida*, is an introduced and invasive species, but can be seasonal in growth even though forming kelp forest. We need to discuss if all habitat forming, or only native species, and if only species with individuals living several years (i.e., not annual), will be included in our results. Thus, a new task for MPA Europe will be to more objectively define species that form significant habitat, such as contributing to “biomes” (large areas occupied by photosynthetic species similar in life-form that provide enduring macroecological habitat for many other species), and place these within the context of the EUNIS classification.

In the next phase of the project, efforts will be directed towards refining the framework used to generate the SDM that underpin the habitat maps. The aim is to optimize model fitting and evaluation processes. Automating the distribution modeling of numerous species presents various challenges when it comes to assessing the biological significance and reliability of the models. To address this, we are in the process of developing pipelines that will enable us to evaluate model quality beyond relying solely on accuracy metrics like AUC and CBI. Moreover, we acknowledge the sensitivity of presence-only models to spatial bias within the occurrence records (e.g. Baker et al., 2022). Therefore, we are implementing methods to effectively control and mitigate spatial bias in the subsequent steps of our modeling.

Following the production of the updated individual models, and the more complete habitat range maps, results will be compared to the biogenic habitat maps available on EMODnet. This comparison will enable to highlight areas of agreement between maps, what may indicate a higher confidence on predictions. This will provide valuable information to guide conservation and monitoring actions to reduce the current pace of biodiversity loss.

Table 2. Area, in km<sup>2</sup>, of biogenic habitats according to the models of habitat distribution. Habitat distribution maps were produced by aggregating individual species distribution models of habitat-forming species obtained through the Maxent algorithm. Models were predicted to the current and future period, in five distinct scenarios (SSP1, SSP2, SSP3, SSP4, and SSP5) and two time periods (2050 and 2100). For the future, the difference of covered area in percentage relative to the current area is shown.

| Biogenic habitat                  | Current area, km <sup>2</sup> | Future area (difference relative to current period, in %) |      |      |      |      |      |      |      |      |      |
|-----------------------------------|-------------------------------|-----------------------------------------------------------|------|------|------|------|------|------|------|------|------|
|                                   |                               | 2050                                                      |      |      |      |      | 2100 |      |      |      |      |
|                                   |                               | SSP1                                                      | SSP2 | SSP3 | SSP4 | SSP5 | SSP1 | SSP2 | SSP3 | SSP4 | SSP5 |
| <b>Cold-water corals reefs</b>    | 5,475,264                     | 104                                                       | 107  | 108  | 109  | 108  | 105  | 113  | 117  | 119  | 117  |
| <b>Bryozoan reefs</b>             | 1,451,134                     | 113                                                       | 117  | 126  | 119  | 131  | 114  | 131  | 161  | 142  | 179  |
| <b>Coralligenous platforms</b>    | 1,189,253                     | 152                                                       | 156  | 155  | 158  | 159  | 152  | 159  | 158  | 155  | 159  |
| <b>Polychaete reefs</b>           | 2,134,427                     | 95                                                        | 95   | 96   | 94   | 98   | 95   | 94   | 103  | 96   | 114  |
| <b>Deep-sea sponge grounds</b>    | 4,964,001                     | 90                                                        | 90   | 89   | 90   | 89   | 83   | 82   | 79   | 80   | 78   |
| <b>Shallow-water sponge reefs</b> | 469,213                       | 89                                                        | 86   | 86   | 83   | 84   | 90   | 83   | 80   | 75   | 75   |
| <b>Mollusc reefs</b>              | 2,823,670                     | 102                                                       | 105  | 105  | 104  | 108  | 104  | 109  | 114  | 110  | 114  |
| <b>Habitat-forming algae</b>      | 953,824                       | 111                                                       | 114  | 109  | 111  | 113  | 109  | 113  | 112  | 109  | 113  |
| <b>Seagrass meadows</b>           | 944,288                       | 91                                                        | 97   | 95   | 91   | 94   | 102  | 105  | 99   | 105  | 100  |



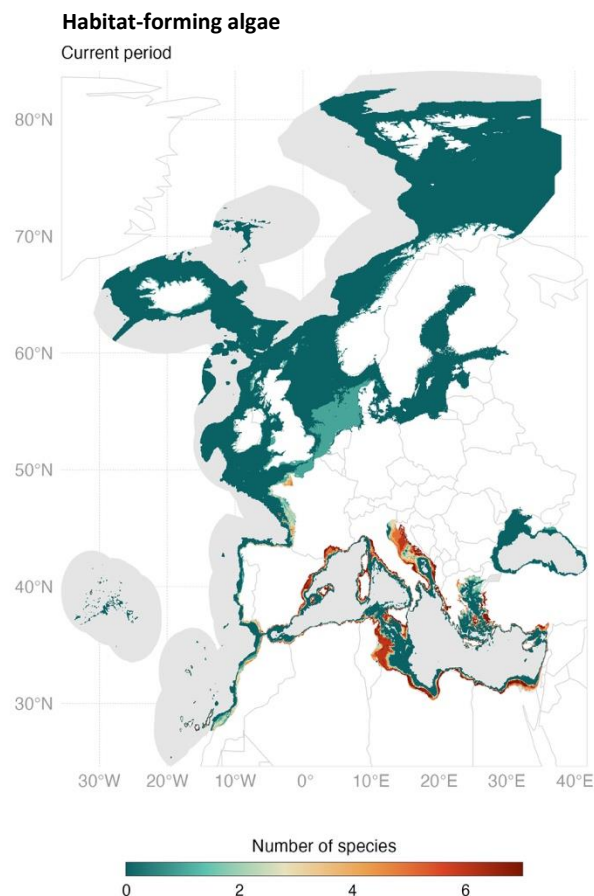


Figure 1. Predicted distribution of habitat-forming macroalgae in the current period according to species distribution models. Individual distribution maps (7 species) were summed to produce the prediction. Areas in light gray depict regions without predictions. See Table 1 for further details.

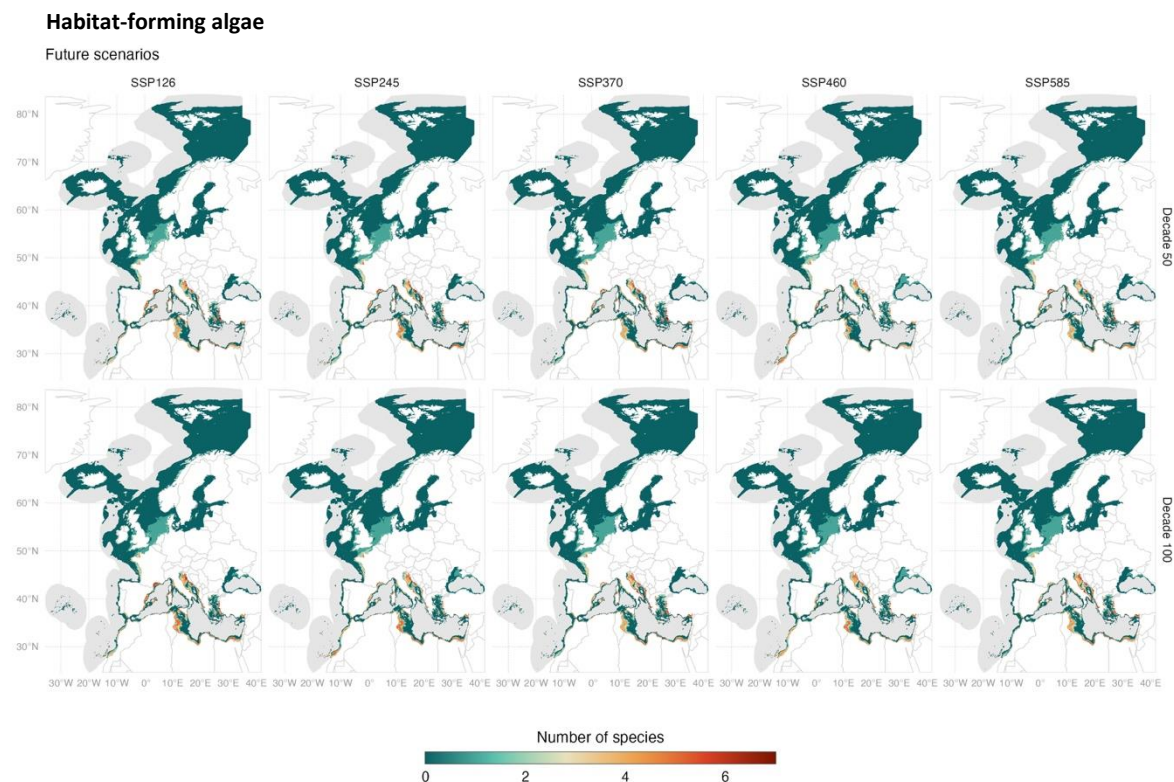


Figure 2. Predicted distribution of habitat-forming macroalgae in the future period according to species distribution models, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 – marked as decade 50, and 2100 – marked as decade 100). Individual distribution maps (7 species) predicted to each scenario/time period were summed to produce the predictions. Areas in light gray depict regions without predictions.

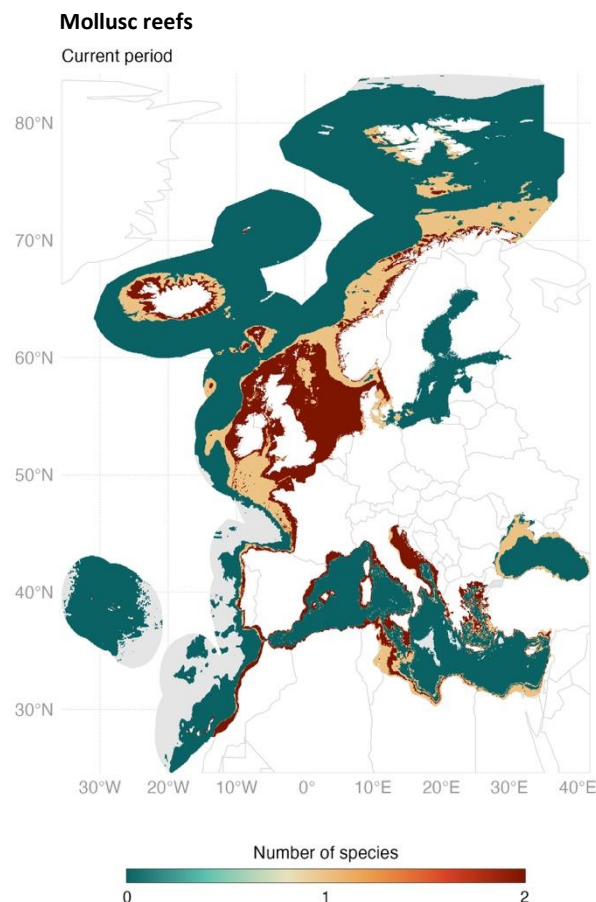


Figure 3. Predicted distribution of mollusc (*Mollusca*) reefs in the current period according to species distribution models. Individual distribution maps (2 species) were summed to produce the prediction. Areas in light gray depict regions without predictions. See Table 1 for further details.

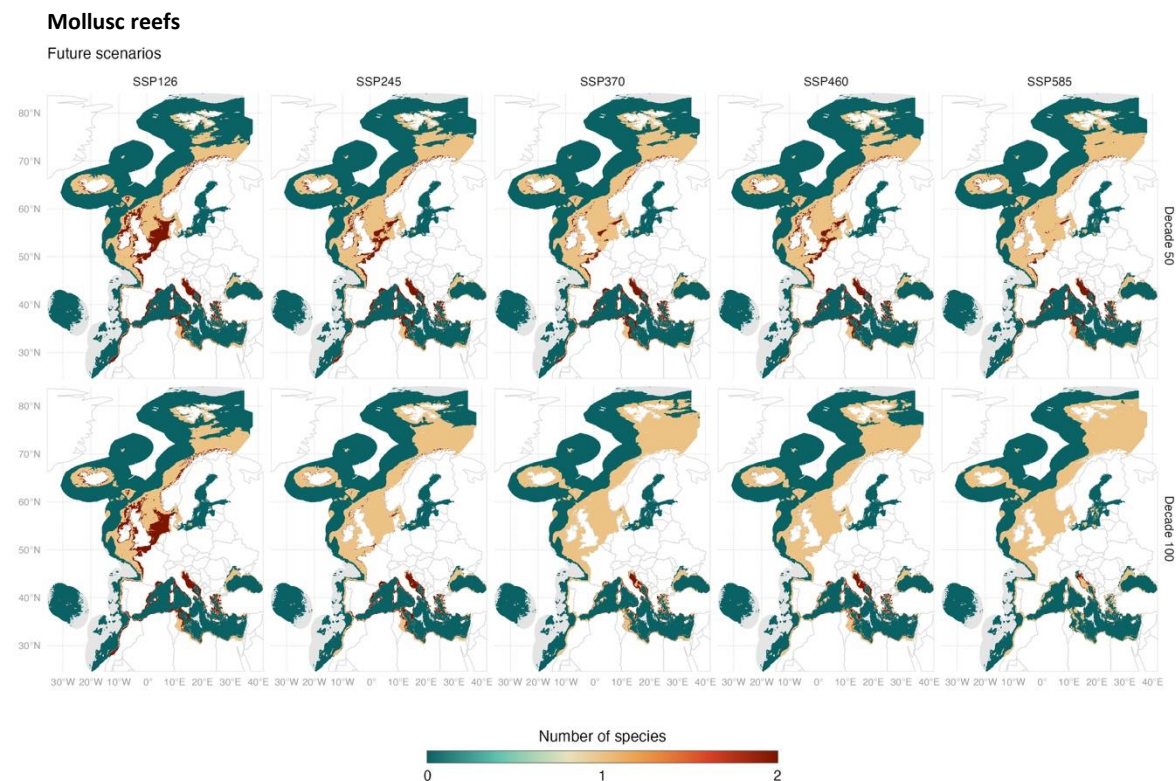
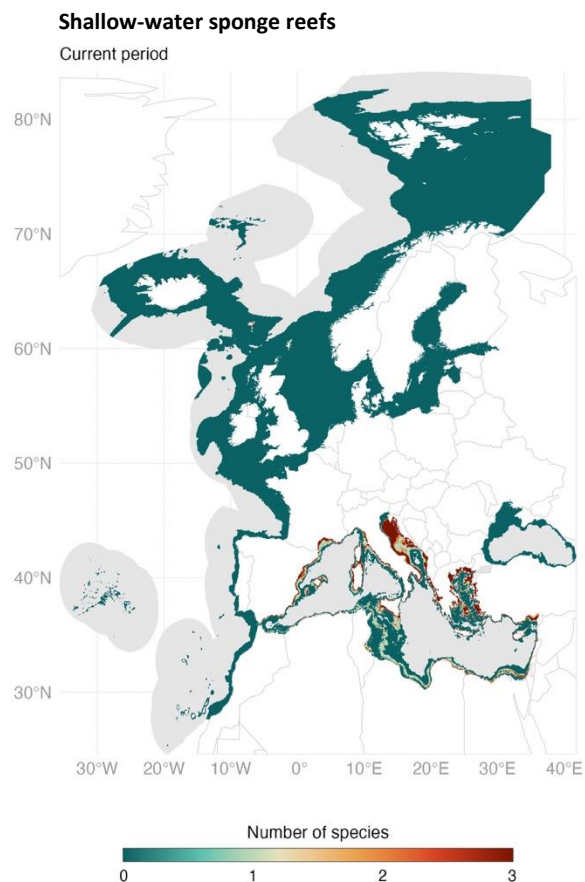
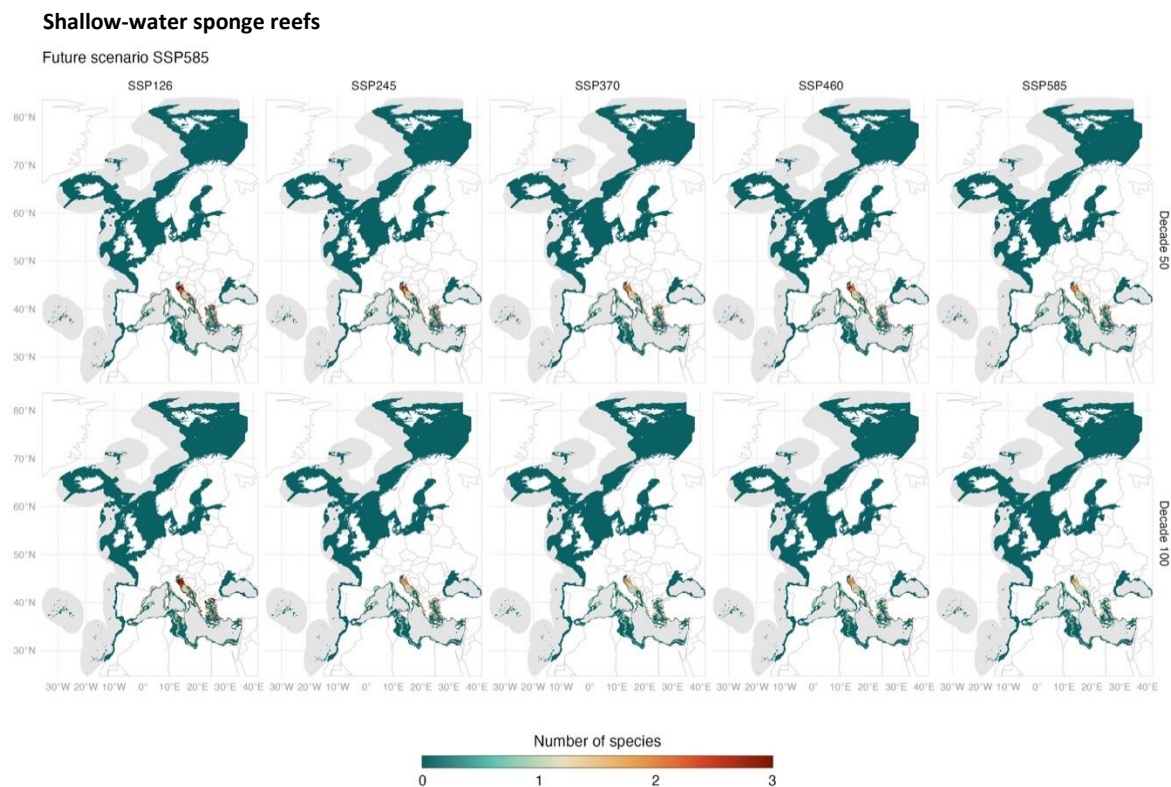


Figure 4. Predicted distribution of mollusc (*Mollusca*) reefs in the future period according to species distribution models, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 – marked as decade 50, and 2100 – marked as decade 100). Individual distribution maps (2 species) predicted to each scenario/time period were summed to produce the predictions. Areas in light gray depict regions without predictions.





*Figure 5. Predicted distribution of shallow-water sponge (Porifera) reefs in the current period according to species distribution models. Individual distribution maps (3 species) were summed to produce the prediction. Areas in light gray depict regions without predictions. See Table 1 for further details.*



*Figure 6. Predicted distribution of shallow-water sponge (Porifera) reefs in the future period according to species distribution models, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 – marked as decade 50, and 2100 – marked as decade 100). Individual distribution maps (3 species) predicted to each scenario/time period were summed to produce the predictions. Areas in light gray depict regions without predictions.*

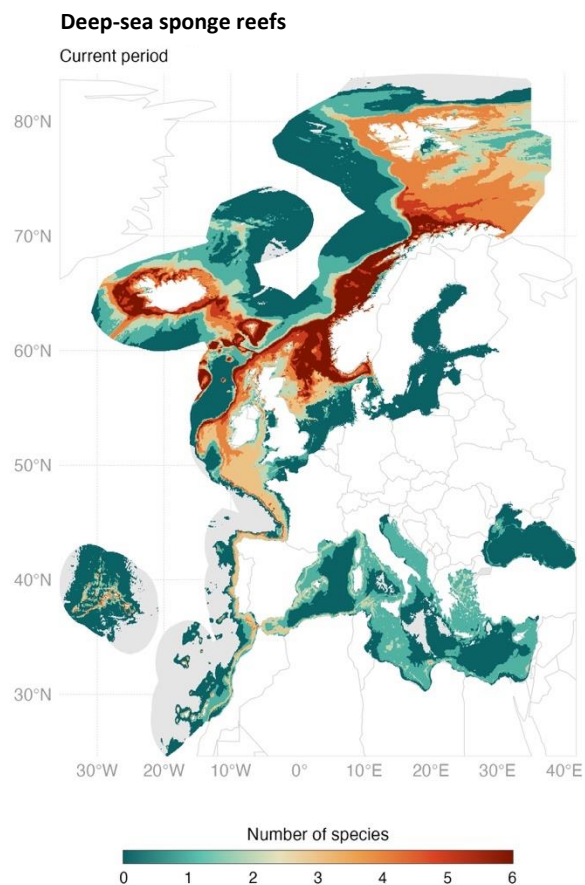


Figure 7. Predicted distribution of deep-sea sponge (*Porifera*) grounds in the current period according to species distribution models. Individual distribution maps (6 species) were summed to produce the prediction. Areas in light gray depict regions without predictions. See Table 1 for further details.

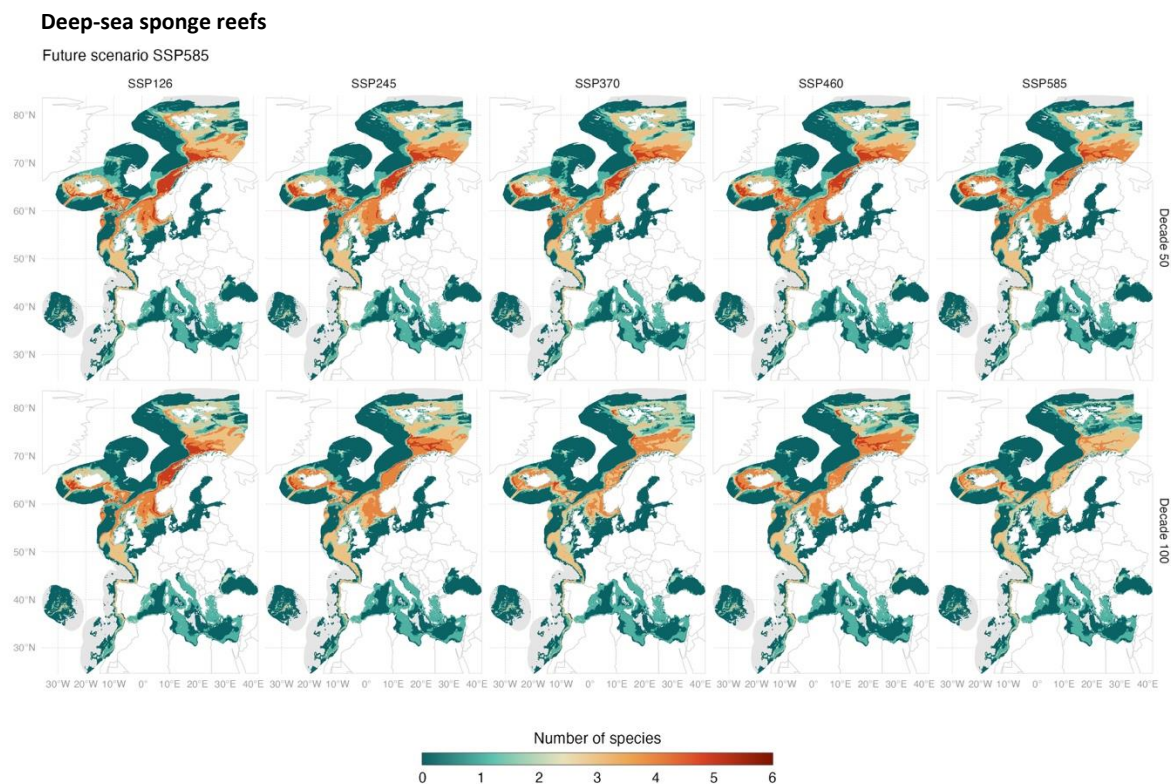


Figure 8. Predicted distribution of deep-sea sponge (*Porifera*) grounds in the future period according to species distribution models, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 – marked as decade 50, and 2100 – marked as decade 100). Individual distribution maps (6 species) predicted to each scenario/time period were summed to produce the predictions. Areas in light gray depict regions without predictions.



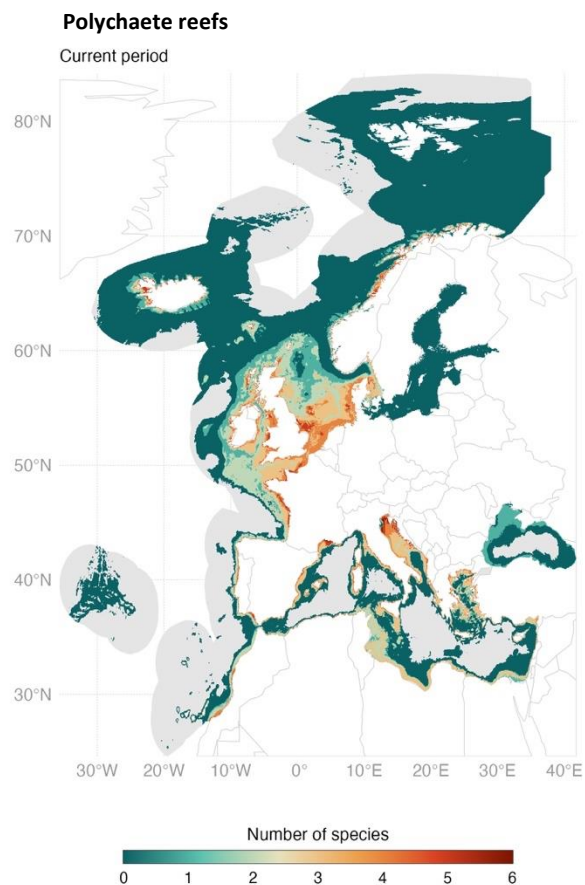


Figure 9. Predicted distribution of polychaete (*Annelida*) reefs in the current period according to species distribution models. Individual distribution maps (6 species) were summed to produce the prediction. Areas in light gray depict regions without predictions. See Table 1 for further details.

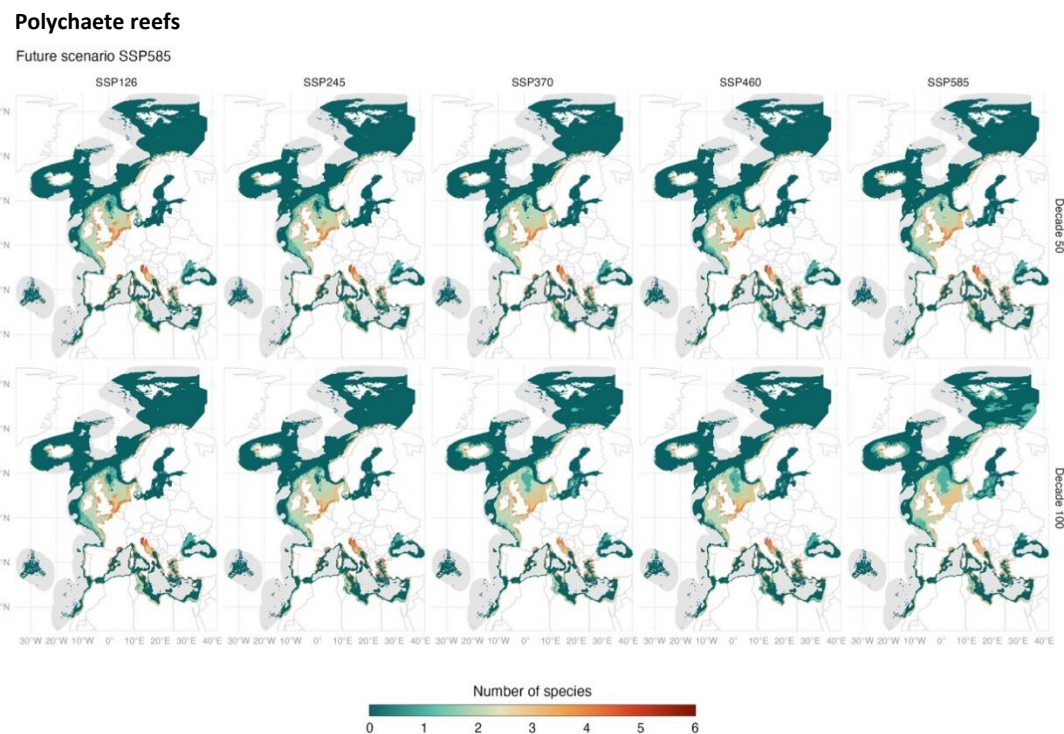


Figure 10. Predicted distribution of polychaete (*Annelida*) reefs in the future period according to species distribution models, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 – marked as decade 50, and 2100 – marked as decade 100). Individual distribution maps (6 species) predicted to each scenario/time period were summed to produce the predictions. Areas in light gray depict regions without predictions.



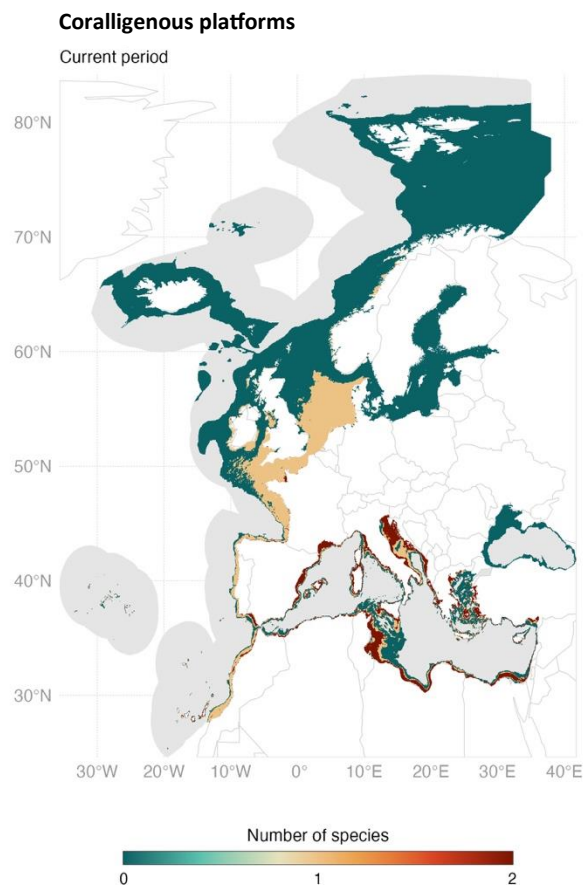


Figure 11. Predicted distribution of coralligenous platforms (formed by species of Rhodophyta: Corallinales) in the current period according to species distribution models. Individual distribution maps (2 species) were summed to produce the prediction. Areas in light gray depict regions without predictions. See Table 1 for further details.

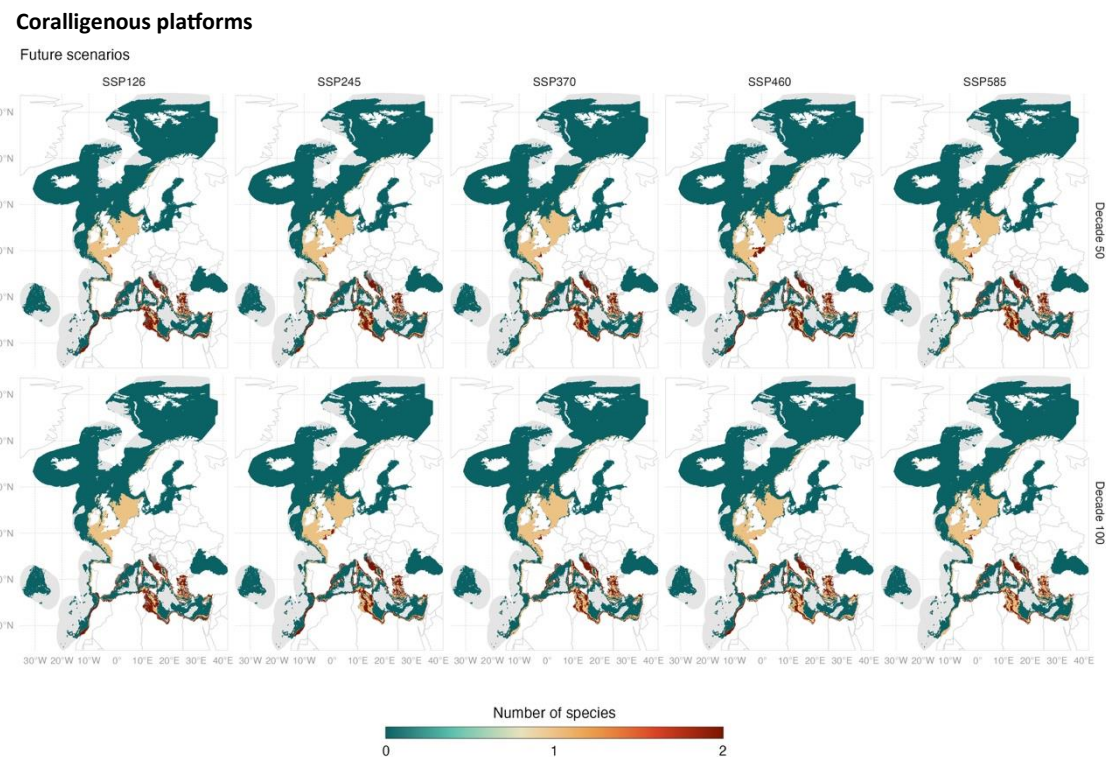


Figure 12. Predicted distribution of coralligenous platforms (formed by species of Rhodophyta: Corallinales) in the future period according to species distribution models, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 – marked as decade 50, and 2100 – marked as decade 100). Individual distribution maps (4 species) predicted to each scenario/time period were summed to produce the predictions. Areas in light gray depict regions without predictions.

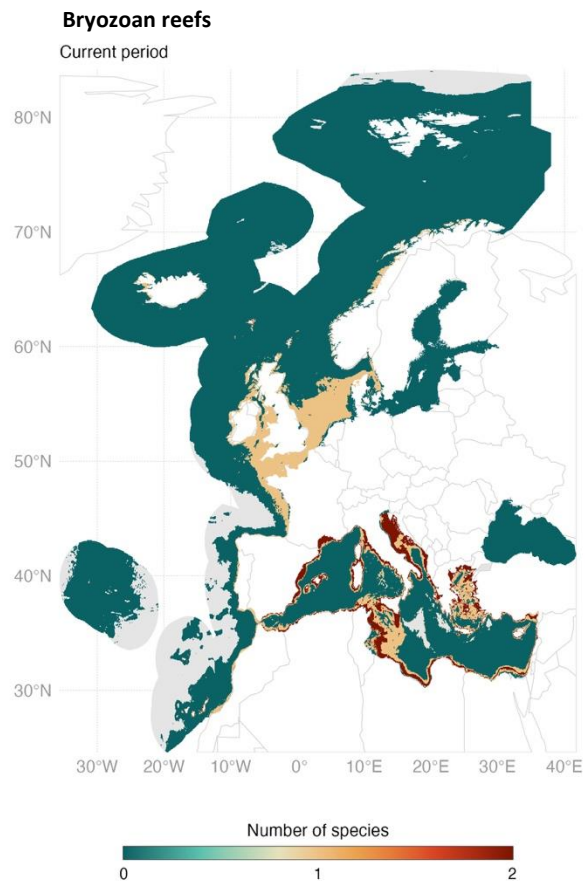


Figure 13. Predicted distribution of bryozoan reefs in the current period according to species distribution models. Individual distribution maps (2 species) were summed to produce the prediction. Areas in light gray depict regions without predictions. See Table 1 for further details.

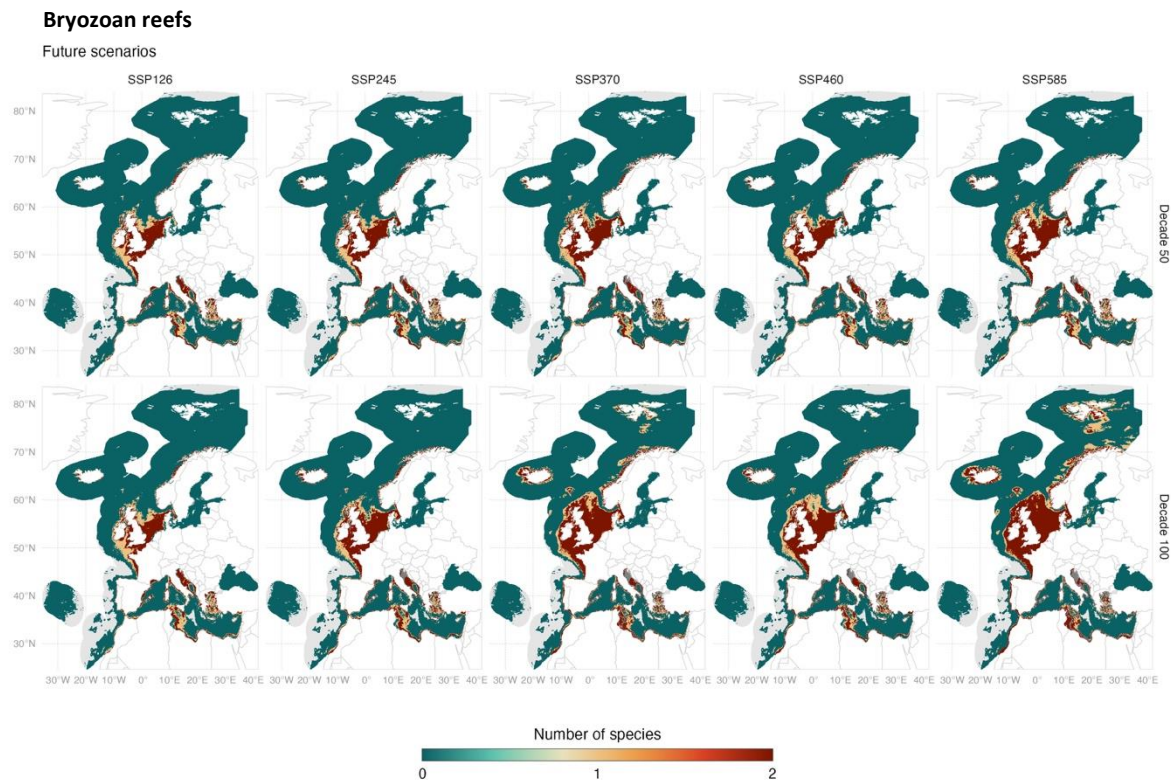


Figure 14. Predicted distribution of bryozoan reefs in the future period according to species distribution models, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 – marked as decade 50, and 2100 – marked as decade 100). Individual distribution maps (3 species) predicted to each scenario/time period were summed to produce the predictions. Areas in light gray depict regions without predictions.



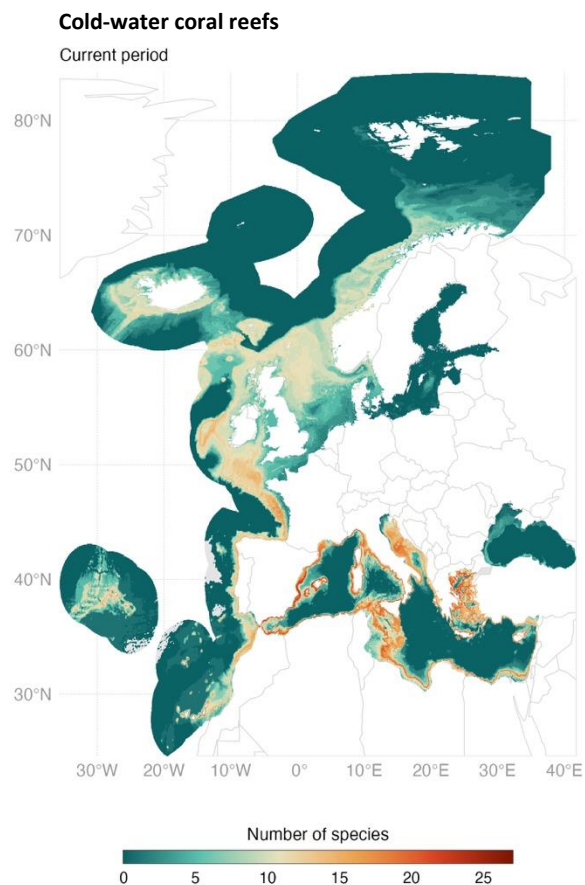


Figure 15. Predicted distribution of cold-water coral reefs in the current period according to species distribution models. Individual distribution maps (31 species) were summed to produce the prediction. Areas in light gray depict regions without predictions. See Table 1 for further details.

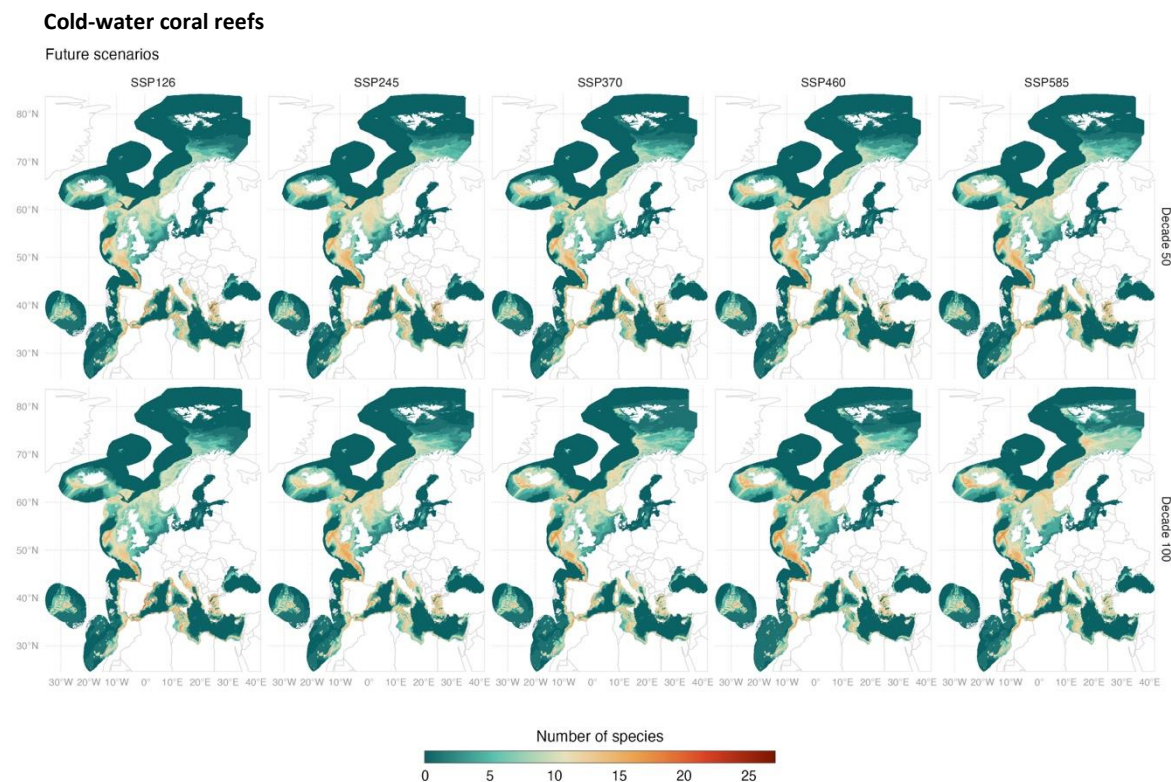


Figure 16. Predicted distribution of cold-water coral reefs in the future period according to species distribution models, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 – marked as decade 50, and 2100 – marked as decade 100). Individual distribution maps (31 species) predicted to each scenario/time period were summed to produce the predictions. Areas in light gray depict regions without predictions.

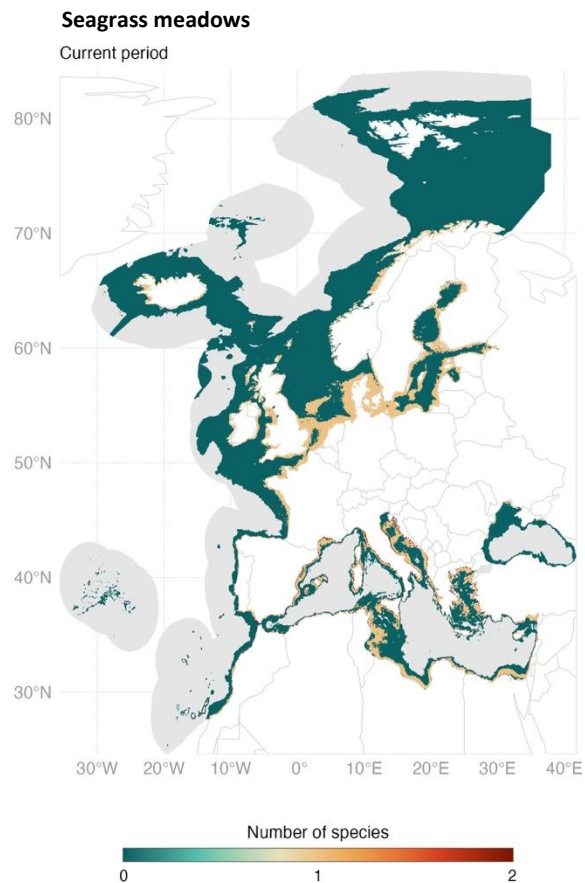


Figure 17. Predicted distribution of seagrass meadows in the current period according to species distribution models. Individual distribution maps (2 species) were summed to produce the prediction. Areas in light gray depict regions without predictions. See Table 1 for further details.

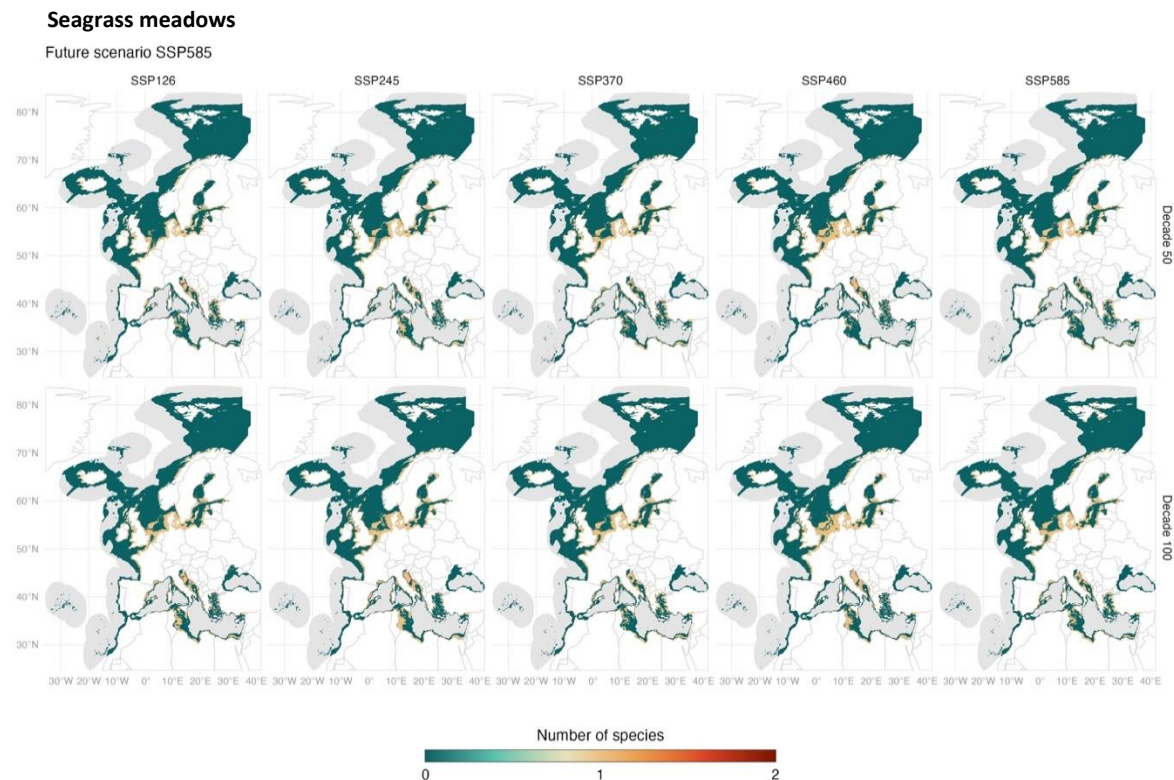


Figure 18. Predicted distribution of seagrass meadows in the future period according to species distribution models, considering five climate scenarios (SSP1, SSP2, SSP3, SSP4 and SSP5) and two time periods (2050 – marked as decade 50, and 2100 – marked as decade 100). Individual distribution maps (2 species) predicted to each scenario/time period were summed to produce the predictions. Areas in light gray depict regions without predictions.



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